Business Applications of Blockchain

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Blockchain: Foundations and Business Value

What Is Blockchain?

Core Definition

- **Distributed ledger**: records shared and maintained by a community; no single party controls it
- **Cryptographic security**: each block links to the previous via a hash; tampering breaks the chain
- **Consensus**: nodes agree on valid transactions via PoW, PoS, or BFT variants
- **Immutability**: altering a confirmed block requires re-computing all subsequent blocks — prohibitively costly

Why It Matters for Business

- **Trust without intermediaries**: math replaces banks, notaries, and clearinghouses
- **Programmable**: smart contracts execute automatically when conditions are met
- **Transparent**: Ethereum's full history is publicly queryable at any time
- **Real-time**: transactions propagate in seconds; finality in minutes

Key Lesson

Blockchain replaces institutional trust with computational trust — value comes from disintermediation, not just digitization.

The Fourth Trust Revolution

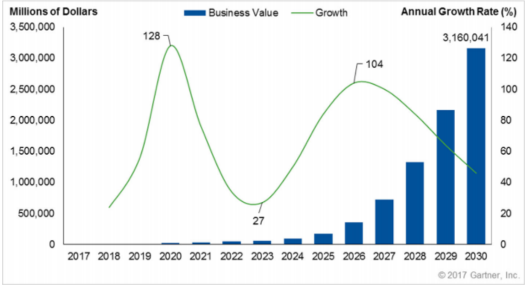
Oral contracts → community witnesses; worked at small scale only

Written contracts → legal systems enforce; requires courts and lawyers

Internet → revolutionized communication but NOT trust; still needs banks and governments online

Blockchain → embeds trust in code; value transfers without needing a trusted third party

Global blockchain market forecast: $1.4 trillion by 2030 (Allied Market Research, 2024)



Source: Gartner (March 2017)

Public, Permissioned, and Private Blockchains

Public Blockchains

- **Open to all**: anyone can read, write, validate (Bitcoin, Ethereum)
- **Decentralized cooperation**: 20,000+ reachable Bitcoin nodes globally
- **Consensus**: Proof-of-Work or Proof-of-Stake
- **Trade-off**: slower, public data, higher energy use

Private Blockchains

- **Single-entity write**: one organization controls the chain
- **Use case**: internal audit trails, enterprise records

Permissioned / Consortium

- **Vetted validators**: only approved nodes participate (banks, insurers)
- **Better performance**: faster finality, lower energy vs. PoW
- **Privacy**: data visible only to members
- **Examples**: Hyperledger Fabric, R3 Corda, JPMorgan Quorum

Key Lesson

Enterprise blockchain is almost always permissioned: privacy, speed, and governance matter more than decentralization for financial institutions.

Enterprise Blockchain Platforms

Hyperledger Fabric (Linux Foundation): modular permissioned platform; IBM leads; used by Walmart Food Trust and Maersk; 300+ enterprise deployments

R3 Corda: purpose-built for financial services; 300+ member institutions; processes derivatives, repo, and trade finance on Corda network

Quorum (JPMorgan/ConsenSys): Ethereum-based permissioned chain; private transactions; powers JPMorgan Onyx network used by 400+ institutions

Ethereum Enterprise Alliance: 250+ members including Microsoft, Mastercard, Deloitte

R3 Corda and Hyperledger: Industry Adoption

R3 Corda: started 2015 by consortium of global banks; open-source DLT tailored to financial contracts; available on Amazon AWS

Members: Barclays, HSBC, BNP Paribas, Deutsche Bank; Goldman Sachs and JPMorgan later exited to build own solutions

Hyperledger: Linux Foundation umbrella; IBM developed Fabric; Intel developed Sawtooth; Quilt handles interledger protocol

Real deployment: IBM Food Trust on Fabric traces food supply for Walmart, Carrefour, and Nestle

Blockchain Business Value: Three Levels

Level 1: Record-Keeping

Tamper-proof records: land registries (Georgia/Bitfury), medical records, diplomas

Who benefits: governments, universities, insurers

Value: eliminates forgery; reduces admin costs

Level 2: Asset Exchange

Tokenized transfer: ownership of digital or real-world assets

Examples: Bitcoin ($277B/day equities settlement equivalent), tokenized bonds

Value: 24/7 settlement, no custodian needed

Level 3: Smart Contracts

Self-executing code: logic runs automatically when conditions are met

Finance: automatic bond coupon payment, parametric insurance payout upon hurricane landfall

Supply chain: supplier paid upon verified delivery via IoT sensor

Key Lesson

Most enterprise blockchain value today is Level 1–2. Level 3 (smart contracts) is the frontier with highest disruption potential.

Blockchain Challenges: What Could Go Wrong

Technical Barriers

- **Scalability**: Bitcoin processes 7 tx/sec vs. Visa's 24,000; Ethereum post-merge handles ~15 tx/sec
- **Interoperability**: hundreds of special-purpose chains cannot yet communicate natively
- **Speed vs. decentralization**: distributed consensus is inherently slower than centralized databases

Governance Failures

- **DAO governance attack**: Beanstalk protocol lost $182M in 2022 via flash-loan governance exploit
- **Open governance**: decentralization does not automatically produce fair or equitable outcomes

Commercial and Regulatory

- **Standards gap**: multiple blockchains need common data and messaging standards to interoperate
- **Commercial conflicts**: competitors must cooperate to achieve network effects — hard in practice (TradeLens shut down 2022)
- **Privacy**: permissioned ledgers expose counterparty data within the consortium
- **Regulatory uncertainty**: EU MiCA (2024) is first comprehensive framework; US still fragmented between SEC and CFTC

Decentralized Finance (DeFi)

DeFi: Finance Without Intermediaries

What Is DeFi?

Definition: financial services built on public blockchains (mainly Ethereum) using smart contracts — no banks, no brokers

Total Value Locked (TVL): peaked at \$180B in November 2021; dropped to \$40B in 2023; recovering to ~\$90B in 2024

Key protocols: Uniswap (\$1.5T+ lifetime trading volume), Aave (\$5B TVL), Compound, MakerDAO (issuer of DAI stablecoin)

Yield farming: users earn token rewards by providing liquidity — annual yields reached 100–1000% at peak (2021)

How AMMs Work

Automated Market Maker (AMM): replaces order books with liquidity pools

Constant product formula: $x \cdot y = k$; price adjusts automatically as traders buy/sell

Uniswap: any user can become a liquidity provider and earn 0.3% fee per trade

No counterparty needed: trades execute against the pool, not against another human

Key Lesson

DeFi eliminates the intermediary but introduces smart contract risk: \$3.8B stolen in DeFi hacks in 2022 alone.

DeFi Protocols: Lending, Borrowing, and Stablecoins

Lending and Borrowing

Aave: largest DeFi lender; $5B TVL; users deposit crypto as collateral and borrow algorithmically; interest rates adjust in real time with supply and demand

Compound: pioneered "money markets" on Ethereum; distributes COMP governance tokens to users

Over-collateralized: borrowers must post 150%+ collateral — protects lenders but limits capital efficiency

Flash loans: borrow millions with no collateral if repaid within one transaction block; used for arbitrage and exploits

Stablecoins and MakerDAO

MakerDAO: issues DAI, a decentralized stablecoin pegged to USD; backed by crypto collateral held in smart contracts

DAI in circulation: ~$5B (2024); used across DeFi as stable unit of account

USDC and USDT: centralized stablecoins; USDC backed 1:1 by US Treasuries; $35B+ circulation

Risk: Terra/LUNA algorithmic stablecoin collapsed May 2022 — $40B market cap wiped out in 72 hours

DeFi Risks and the Hack Problem

Smart Contract Vulnerabilities

Scale of losses: \$3.8B stolen from DeFi protocols in 2022 (Chainalysis); \$1.7B in 2023

Ronin Bridge hack: \$625M stolen from Axie Infinity's Ethereum bridge (March 2022) — largest DeFi hack

Reentrancy attacks: code calls external contract before updating its own state; The DAO hack (2016) lost \$60M, triggered Ethereum hard fork

Oracle manipulation: attackers distort price feeds to drain liquidity pools

Governance and Regulatory Risk

Beanstalk attack: attacker borrowed \$1B via flash loan to gain majority voting power; passed malicious governance proposal; stole \$182M — all in one transaction

Regulatory pressure: US SEC charged Uniswap Labs (2024); EU MiCA imposes compliance on DeFi front-ends

No recourse: unlike bank deposits, DeFi losses are not insured or reversible

Key Lesson

Code is law in DeFi — but bugs in the code can be irreversible. Audit costs and formal verification are the new compliance.

NFTs and Digital Ownership

NFTs: From Hype to Real Use Cases

What Are NFTs?

- **Non-Fungible Token**: unique cryptographic token on a blockchain proving ownership of a specific digital (or physical) asset
- **Peak market**: $25B in trading volume in 2021; collapsed 97% by 2023 — most speculative NFTs worth near zero
- **Why most failed**: pure speculation with no utility; buyers could not sell; platforms lost liquidity
- **ERC-721 standard**: Ethereum token standard enabling unique, non-interchangeable digital assets

Real Use Cases Beyond Art

- **Nike**: generated $185M+ in NFT/digital sneaker revenue via .SWOOSH platform on Polygon; smart contracts auto-pay royalties to co-creators
- **Gaming**: Axie Infinity peaked at 2M daily players; NFT characters tradeable; Ronin network processed $4B in NFT transactions (2021)
- **Ticketing**: AXS and GET Protocol use NFTs to prevent scalping; ticket history on-chain; resale royalties flow to artists
- **Starbucks Odyssey**: blockchain loyalty stamps (NFTs) on Polygon; earn real-world rewards

NFT and Digital Identity: Self-Sovereign Identity

The Identity Problem

- **Fragmented today**: identity controlled by governments, Google, Facebook, banks — each holds a piece
- **Breach risk**: Equifax 2017 exposed 147M Americans; single point of failure = mass identity theft
- **Privacy erosion**: centralized platforms profile and monetize identity data without consent
- **Exclusion**: 1.4B adults lack official ID and cannot access banking or healthcare

Decentralized Identity (DID)

- **DIDs**: globally unique identifiers on blockchain; user-controlled, not platform-controlled
- **Verifiable Credentials**: signed proofs (diploma, license, age) presented without revealing full identity
- **Microsoft Entra Verified ID**: enterprise DID on Ethereum (2023)
- **EU Digital Identity Wallet** (2024): citizens store national IDs in blockchain-secured wallet

Key Lesson

NFTs are the ownership primitive; DIDs are the identity primitive — together they enable a user-controlled digital world.

Tokenization of Real-World Assets

Tokenization: Institutional Finance Goes On-Chain

What Is Tokenization?

- **Definition**: ownership of a real-world asset (bond, equity, real estate, fund) as a digital token on a blockchain
- **Why now**: blockchain settlement is 24/7, instant, programmable; traditional T+2 settlement is closed on weekends
- **Fractional ownership**: a $10M building divided into 10,000 tokens of $1,000 each — democratizing institutional assets
- **Market size**: RWA exceeded $8B in 2024; projected $16T by 2030 (BCG)

Who Is Doing It?

- **BlackRock BUIDL**: tokenized money market fund on Ethereum; $500M+ AUM weeks after launch (2024); daily dividends on-chain
- **JPMorgan Onyx**: $700B+ tokenized repo transactions; intraday settlement in minutes
- **Franklin Templeton FOBXX**: $1B+ US Treasuries on Stellar blockchain
- **Singapore Project Guardian**: MAS-led trial tokenizing bonds and FX with JPMorgan, DBS, SBI

Key Lesson

BlackRock CEO Larry Fink called tokenization “the next generation for markets.” This is not speculation — it is institutional infrastructure.

Smart Contracts in Finance: Automating Obligations

Bond and Derivatives Settlement

Automated bond coupons: smart contract releases coupon payment to all holders on the exact due date — no bank transfer, no delay

ISDA CDM (Common Domain Model): industry standard for representing derivatives contracts as machine-readable code; 30+ major banks participating

Daimler bond (2017): first blockchain-based corporate bond issuance using Hyperledger; UBS led with Barclays, Credit Suisse, and others

ASX: replaced CHESS clearing system with DLT-based post-trade infrastructure (long-running project, delayed to 2025)

Parametric Insurance

How it works: policy terms encoded in smart contract; oracle (e.g., weather data feed) triggers automatic payout when condition met

Hurricane example: if Category 3+ hurricane makes landfall within 50 miles of insured property → smart contract pays within minutes, no claims adjuster needed

Etherisc: decentralized insurance protocol; flight delay insurance pays automatically based on public flight data

AXA Fizzy: parametric flight delay product on Ethereum (2017–2019); proof of concept for mainstream insurers

Supply Chain and Trade Finance

Blockchain in Supply Chain: The Food Safety Case

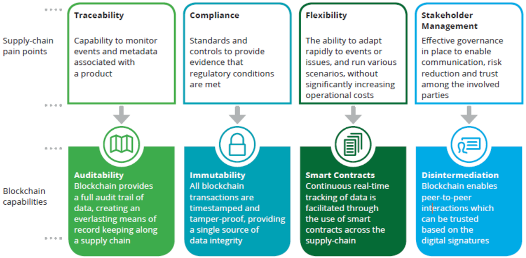
IBM Food Trust (on Hyperledger Fabric): Walmart, Carrefour, and Nestle trace food from farm to shelf

Speed: traces food contamination source in **2.2 seconds** vs. 7 days using paper records — critical for recalls

Walmart mandate: all leafy green suppliers must upload data to IBM Food Trust; 2019 deadline

De Beers Tracr: tracks 25%+ of global diamond supply on blockchain; every stone has verified conflict-free record; launched commercially 2022

DHL and Accenture PharmaChain: tracks temperature-sensitive drugs; reduces counterfeit pharmaceutical risk



TradeLens: Lessons from a Blockchain Shutdown

What TradeLens Was

- **Maersk and IBM** joint venture (2018–2022); digitized shipping documentation across 100+ ports
- **Ambition**: eliminate paper bills of lading; connect shippers, ports, customs authorities on one ledger
- **Scale**: 300+ ocean carriers and logistics partners; billions of shipping events recorded
- **Claimed savings**: 40% reduction in documentation processing time

Why It Failed (2022)

- **Adoption problem**: competitors (CMA CGM, MSC) refused to share data on a Maersk-controlled platform
- **Commercial conflict**: network effects require universal participation; fragmentation prevented critical mass
- **The lesson**: blockchain technology worked; business model and governance failed
- **Implication**: supply chain blockchains need neutral governance (Hyperledger, not competitor-led)

Key Lesson

TradeLens is the canonical lesson: blockchain adoption fails not from technology but from governance and commercial incentives.

Trade Finance: The $18 Trillion Gap

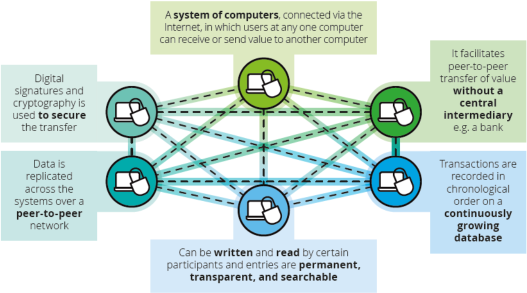
Trade finance gap: $18T in global unmet trade finance demand (ADB, 2023); SMEs hardest hit due to documentation burden

Current process: letters of credit, bills of lading, certificates of origin — mostly still paper; takes 5–10 days

We.Trade: European bank consortium (HSBC, Deutsche Bank, Rabobank) using Hyperledger for open account trade finance

Marco Polo: R3 Corda-based trade finance network; automates payment obligations when goods confirmed delivered

Contour: formerly Voltron; digitizes letters of credit; reduced processing from 5–10 days to under 24 hours



Source: Deloitte (2017)

Blockchain and Contractual Incompleteness

Hold-up problem: when investments are relationship-specific, one party seeks to renegotiate ex post — leads to underinvestment (Williamson 1975; Klein, Crawford, and Alchian 1978)

Traditional solutions: vertical integration, long-term contracts, property rights (Grossman and Hart 1986)

Blockchain solution: smart contracts enforce payment automatically upon verified delivery; immutable records reduce information asymmetry; decentralized verification lowers trust costs

Research (Chen, Hu, Wang, Wu): exploit staggered U.S. state pro-blockchain laws (2015–2019) as quasi-natural experiment

Ernst & Young Unveils Supply Chain Manager on Polygon Network
CoinDesk, May 17, 2022

US Air Force Pumps $30M Into Blockchain for Supply Chains
Blockworks.co/news
February 1, 2023

UNILEVER UNVEILS BLOCKCHAIN SUPPLY CHAIN TRACKING SOLUTION TO INCREASE PALM OIL TRANSPARENCY
GLOBAL COSMETICS NEWS, MAR 24, 2022

Samsung Joins Blockchain Bandwagon to Manage its Supply Chain
HT Tech, Aug. 19, 2022

MAJOR SEAFOOD PRODUCER THAI UNION LAUNCHES SHRIMP BLOCKCHAIN TRACEABILITY PILOT
Ledger Insights, February 24, 2022

Californian Cannabis Nursery Adopts Blockchain to Keep Record of Weed Strains
Investing.com, Jan 16, 2023

Research Evidence: Blockchain and Firm Value

Empirical Strategy

Asset Specificity Index (ASI): text-based measure of each firm's exposure to contractual incompleteness — how relationship-specific are its supply chain investments?

Identification: staggered triple-differences (DDD); exploits which U.S. state passed a pro-blockchain law and when (2015–2019)

Outcomes: Tobin's Q, R&D spending, patenting activity, vertical integration, geographic proximity of customers

Key Findings

Firm value: after pro-blockchain law, high-ASI firms show greater Tobin's Q increase than low-ASI firms

Innovation: high-ASI firms increase R&D and blockchain patenting; patents more specific and higher value

Geography: high-ASI firms shift away from geographically proximate customers — blockchain relaxes the proximity constraint for trust

Key Lesson

Blockchain creates measurable economic value by reducing the cost of contractual incompleteness — especially for relationship-specific investments.

Blockchain Legislation and Firm Value

After a state passes a **pro-blockchain law**, high-ASI firms in that state exhibit a **greater increase in Tobin's Q** than low-ASI firms

Effect is economically significant and robust to controlling for state and industry fixed effects

Mechanism: blockchain reduces **hold-up risk** for firms with specific assets, raising expected payoff from relationship-specific investment

Consistent with Grossman-Hart property rights theory: improving contracting technology increases investment and firm value

	Tobin's Q (1)	*Tobin's Q Alt.* (2)	*Tobin's Q, Quarterly* (3)	*Tobin's Q Alt., Quarterly* (4)
Post Treat × ASI	0.154***	0.267***	0.140***	0.283***
	(0.03)	(0.07)	(0.03)	(0.07)
Post Treat	-0.078***	-0.140***	-0.066***	-0.115***
	(0.02)	(0.04)	(0.02)	(0.04)
Treat × ASI	-0.101	-0.152	-0.042	-0.081
	(0.07)	(0.21)	(0.07)	(0.20)
ASI	0.089	0.145	0.027	-0.006
	(0.06)	(0.17)	(0.06)	(0.17)
Size	-0.058***	0.114***	-0.064***	0.129***
	(0.01)	(0.02)	(0.01)	(0.02)
ROA	0.026	0.660***	-0.086**	1.083***
	(0.02)	(0.05)	(0.04)	(0.09)
Cash	0.339***	0.621***	0.293***	0.728***
	(0.03)	(0.07)	(0.02)	(0.07)
Sales	0.144***	-0.052**	0.587***	0.196***
	(0.01)	(0.03)	(0.03)	(0.07)
HHI	-0.041**	-0.008	-0.152**	-0.152
	(0.02)	(0.05)	(0.07)	(0.18)
Industry Intangibles	0.012***	0.031***	0.013***	0.034***
	(0.00)	(0.01)	(0.00)	(0.01)
Industry R&D	-0.000	-0.002	0.002	-0.003
	(0.00)	(0.01)	(0.00)	(0.01)
Firm FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	No	No
Quarter FEs	No	No	Yes	Yes
Observations	33,322	26,463	138,458	72,970
Adj. R-squared	0.766	0.845	0.806	0.866

Blockchain Legislation and Innovation

After a pro-blockchain law, **high-ASI firms** exhibit:

- Greater increase in **R&D expenditure**
- More **blockchain-specific patenting**
- Higher **average patent value** (forward citations)
- More technologically **specific patent claims**

Interpretation: legal certainty around blockchain reduces risk of investing in blockchain-specific innovation

Firms that previously under-invested due to hold-up risk now invest more when contracts become self-enforcing

	R&D Intensity (1)	*All Patents* (2)	*Blockchain Patents* (3)	*Patent Value* (4)	*Patent Generality* (5)
Post Treat × ASI	0.079**	-0.081	0.082***	0.800**	-0.151***
	(0.03)	(0.07)	(0.03)	(0.36)	(0.02)
Post Treat	-0.020	0.045	-0.025**	-0.377*	0.070***
	(0.02)	(0.04)	(0.01)	(0.21)	(0.01)
Treat × ASI	0.184	-0.062	0.039	-0.833	0.082**
	(0.14)	(0.15)	(0.03)	(0.72)	(0.04)
ASI	-0.189	0.137	-0.001	0.171	-0.034
	(0.13)	(0.09)	(0.01)	(0.45)	(0.02)
Controls	Yes	Yes	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes
Observations	19,833	33,381	33,381	33,381	33,381
Adj. R-squared	0.813	0.880	0.380	0.720	0.485

Blockchain, Geography, and Supply Chain Structure

Finding: after a pro-blockchain law, high-ASI firms shift away from **geographically proximate customers**

Mechanism: proximity was a substitute for trust in traditional supply chains; face-to-face monitoring reduced hold-up risk

Blockchain substitutes: immutable records and smart contracts provide trust at a distance — physical proximity is no longer needed for contractual enforcement

Implication: blockchain enables global supply chains for firms that previously had to operate locally due to trust constraints

Panel A: Supplier-Level Tests

	# of Nearby Cust.		*% of Nearby Cust.*		*Keep Any Nearby Cust.*		*Add Any Distant Cust.*	
	< 50 km	< 100 km	< 50 km	< 100 km	< 50 km	< 100 km	≥ 300km	≥ 400 km
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post Treat × ASI	-0.119**	-0.110**	-0.101**	-0.098*	-0.146**	-0.152**	0.214**	0.196*
	(0.05)	(0.05)	(0.05)	(0.05)	(0.06)	(0.06)	(0.10)	(0.10)
Post Treat	0.077***	0.069**	0.071**	0.064*	0.092**	0.094**	-0.114*	-0.114*
	(0.03)	(0.03)	(0.03)	(0.03)	(0.04)	(0.04)	(0.06)	(0.06)
Treat × ASI	0.068	0.097	0.063	0.056	-0.071	-0.079	-0.294	-0.273
	(0.12)	(0.13)	(0.10)	(0.11)	(0.14)	(0.14)	(0.21)	(0.20)
ASI	0.074	0.061	0.002	0.007	0.118	0.149	0.004	0.035
	(0.09)	(0.09)	(0.09)	(0.09)	(0.10)	(0.10)	(0.16)	(0.15)
# of Customers	0.045***	0.052***			0.028***	0.028***	0.111***	0.103***
	(0.01)	(0.01)			(0.01)	(0.01)	(0.01)	(0.01)
	(0.00)	(0.00)	(0.00)	(0.00)	(0.01)	(0.01)	(0.01)	(0.01)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FEs								
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,327	7,327	7,327	7,327	7,200	7,200	7,200	7,200
Adj. R-squared	0.862	0.861	0.863	0.866	0.705	0.706	0.355	0.356

Cross-Border Payments and Remittances

Cross-Border Payments: The Blockchain Opportunity

The Problem with Traditional Payments

SWIFT: 11,000+ member institutions; processes \$5T/day; but settlement takes 1–5 business days and involves 3–5 correspondent banks

Remittance cost: global average fee is 6.3% (World Bank, 2024); Sub-Saharan Africa averages 8%

Philippines corridor: OFW remittances \$36B/year; traditional cost 7%; blockchain alternatives reduce to 2%

SWIFT GPI: launched 2017 to improve speed; now covers 95% of SWIFT traffic; same-day delivery in most cases — but still not real-time

Blockchain Alternatives

Ripple/XRP: real-time gross settlement; \$30B+ market cap; used by Santander, SBI Holdings, American Express for cross-border payments; processes tx in 3–5 seconds at \$0.0001 cost

Stellar/XLM: MoneyGram partnership; remittance between US and Philippines using USDC on Stellar; settles in seconds

JPMorgan Onyx: tokenized deposits for bank-to-bank settlement; 400+ institutions; processes interbank payments in minutes

mBridge: BIS multi-CBDC platform linking China, Hong Kong, Thailand, and UAE; tested \$22M in real transactions (2022 pilot)

Consumer Blockchain Payments: Real Deployments

Bitcoin Lightning Network

Layer-2 solution: payment channels off the main Bitcoin blockchain; settles back to chain periodically

Speed: near-instant; **cost**: fractions of a cent per transaction

El Salvador: legal tender since 2021; Chivo wallet used by 4M+ citizens; remittances from US now 2% cost vs. 7% traditional

Strike: US app using Lightning for instant low-cost cross-border payments; also used by Cash App

B2B and Institutional Payments

JPMorgan Onyx/Liink: interbank information and payment network; 400+ global institutions; reduces settlement from days to minutes

Brave Browser and BAT: 50M+ users earn Basic Attention Tokens for viewing privacy-respecting ads; advertisers get transparent engagement data; redefines digital ad economics

Visa and Mastercard: both now settle crypto transactions using USDC on Ethereum; Visa processed $1B in USDC settlements in 2021

Enterprise Blockchain and Capital Markets

Capital Markets: Blockchain Settlement

The Current System

- **US cash equities**: trades approximately 7 billion shares ($277B notional) per day; multiple intermediaries required: exchanges, brokers, DTC, custody banks
- **T+2 settlement**: takes 2 business days; ties up $50B+ in margin and capital daily
- **Error rate**: about 10% of trades contain errors requiring costly manual correction
- **Back-office cost**: US banks spend $20B+/year on clearing and settlement infrastructure

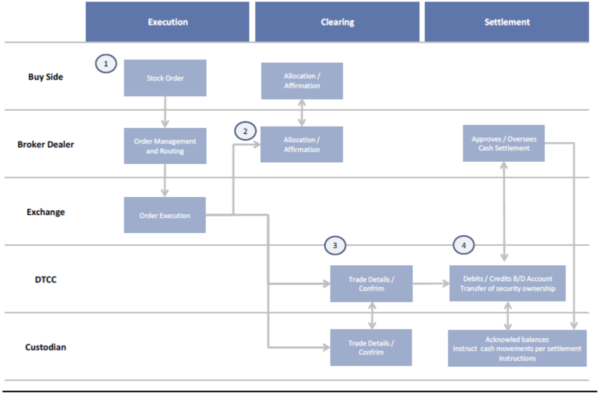
How Blockchain Helps

- **Near real-time settlement**: atomic swap of cash and security simultaneously; no counterparty risk during settlement window
- **Error reduction**: shared ledger eliminates reconciliation between counterparties; no more break reports
- **Estimated savings**: $2B/year in back-office costs alone (Oliver Wyman, 2016); up to $12B globally (Accenture, 2024)
- **T+1 achieved**: SEC moved US markets to T+1 settlement in May 2024 — blockchain technology enabled the infrastructure

Key Lesson

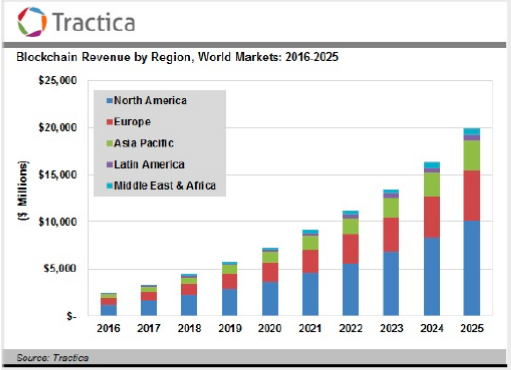
Every day of settlement delay ties up capital. Moving from T+2 to T+0 could free $50B+ in collateral globally — that is real economic value.

How US Equities Are Traded Today



Source: DTCC, Goldman Sachs Global Investment Research.

Enterprise Blockchain: Projected Market Growth



Rapid scaling: enterprise blockchain revenue grew from $2.5B (2016) to $19B (2024); forecast $1.4 trillion business value by 2030

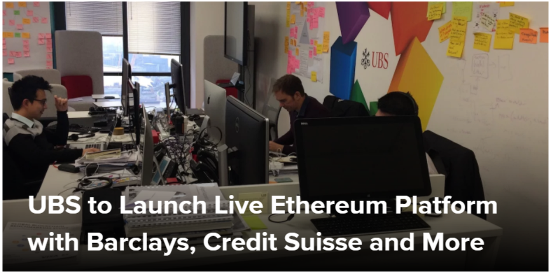
Finance dominates: financial services accounts for 38% of enterprise blockchain spending

80% of global financial institutions have active blockchain initiatives (PwC, 2024)

Supply chain: 60% of supply chain leaders report improved traceability and trust from blockchain adoption

WEF prediction: 10% of global GDP will be stored or transacted on blockchain by 2030

UBS and Daimler: Bond Issuance on Blockchain



UBS initiative (2017): project to make it easier for banks to reconcile counterparty data; partners include Barclays, Credit Suisse, KBC, SIX, and Thomson Reuters

Smart contracts: ensure accuracy of counterparty information in near real-time instead of periodic manual reconciliation

Daimler bond: first corporate bond issued on Hyperledger blockchain; coupon payments automated via smart contract

Impact: demonstrated that blue-chip issuers can use blockchain for primary issuance; reduced documentation cost and settlement risk

Australian Stock Exchange: DLT for Post-Trade

FINANCIAL TIMES

ASX chooses blockchain for equities clearing

Australian bourse becomes first big exchange to commit to the technology

Jamie Smyth in Sydney DECEMBER 6, 2017

The Australian Securities Exchange is planning to use blockchain technology to manage the clearing and settlement of equities.

The decision to replace its ageing clearing and settlement system, known as Chess, with distributed ledger technology — the engine behind the digital currency bitcoin — makes it one of the world's first global exchanges to commit to the technology.

The ASX on Thursday said Digital Asset Holdings, the company run by former JPMorgan Chase executive Blythe Masters, has built and tested the software.

"ASX has been carefully examining distributed ledger technology for almost two and a half years, including the last two years with Digital Asset," said Dominic Stevens, chief executive of ASX. He said the system would cut trading costs and put Australia "at the forefront of innovation in financial markets".

Central Bank Digital Currencies (CBDC)

CBDC: Architecture and Global Race

CBDC Design Choices

Direct model: central bank maintains individual wallets; highest control, highest operational burden

Indirect model: commercial banks issue CBDC-backed tokens; central bank is wholesale layer; most privacy-preserving

Hybrid: central bank backstop; banks handle retail — China e-CNY uses this model

Programmability: spending constraints (welfare CBDC for food only), expiry dates, automatic tax withholding

Global Deployments

China e-CNY: 260M wallets; $250B+ transactions (2024); tested at 2022 Beijing Olympics; programmable with expiry

mBridge (BIS): multi-CBDC platform linking China, Hong Kong, Thailand, UAE; eliminates correspondent banking

EU digital euro: design phase 2024; launch 2027; privacy vs. AML tensions dominate

Bahamas Sand Dollar: world's first live retail CBDC (2020); financial inclusion focus

Key Lesson

130 countries (98% of global GDP) are exploring CBDCs. The question is not if, but when and how.

Government and Public Sector Applications

Government Roles in the Blockchain Economy

Enable and Catalyze

Singapore MAS Project Guardian: tokenized bonds and FX tested with JPMorgan, DBS, and SBI; policy sandbox for institutional DeFi

UAE Blockchain Strategy 2031: target to put 50% of government transactions on blockchain; estimated $3B annual savings

Estonia e-Governance: citizen IDs, health records, property registry, and voting on blockchain; saves estimated 2% of GDP annually in admin costs

Dubai: connects 30+ departments; eliminates 100 million paper documents per year

Regulate and Govern

EU MiCA (2024): world's first comprehensive crypto and digital asset regulatory framework; covers stablecoins, exchanges, and token issuers

US fragmentation: SEC classifies most tokens as securities; CFTC claims Bitcoin and Ethereum as commodities; no unified framework yet

Policy challenge: traditional regulation is too slow for decentralized systems; principle-based frameworks with embedded compliance code ("RegTech") are the frontier

China approach: banned crypto trading; promotes e-CNY and blockchain for supply chain and finance

Blockchain for Digital Identity and Public Records

Land titles: worldwide "dead capital" ≈\$20T — people lack legal title to homes and land (Hernando de Soto)

Georgia/Bitfury: first nation to register land titles on public blockchain (Bitcoin); pilot successful, expanded nationally

Other pilots: Sweden, Honduras, and Cook County (Illinois) testing blockchain land registries

PKI and blockchain: public-private key cryptography is native to blockchain; enables digital signatures without certificate authorities

Korea K-DID: blockchain-based ID integrated into mobile banking and e-government services

STATIC → DYNAMIC

	Rarely changes	Occasionally changes	Frequently changes
INHERENT ATTRIBUTES Attributes intrinsic to an individual are not defined by relationships with other entities.	• Date of birth • Fingerprints/biometrics • Mother/Father • Sibling	• DNA • Hair color • Sex • Spouse	• Height • Weight • Age • Personal health
ACCUMULATED ATTRIBUTES Attributes that are gathered or develop over time	• Health records • Education credentials • Email address	• Vehicle ownership • Personal address • Occupation • Employer	• Preferences & behaviors • Sharing economy ratings • Social media interactions • Wealth/debt
ASSIGNED ATTRIBUTES Attributes tied to relationships with other entities	• Legal Name • Social security number • Driver's license number • Tax ID number	• Credit card number • Insurance policy ID • Nationality • Telephone number	• 2 factor auth. code • Username/password • Loyalty cards • Airline/event tickets

Source: Illinois Blockchain Report

Social Benefits and Smart Government Programs

The challenge: welfare programs (Medicaid, food stamps, student aid) depend on complex eligibility checks across fragmented databases — causing delays and fraud

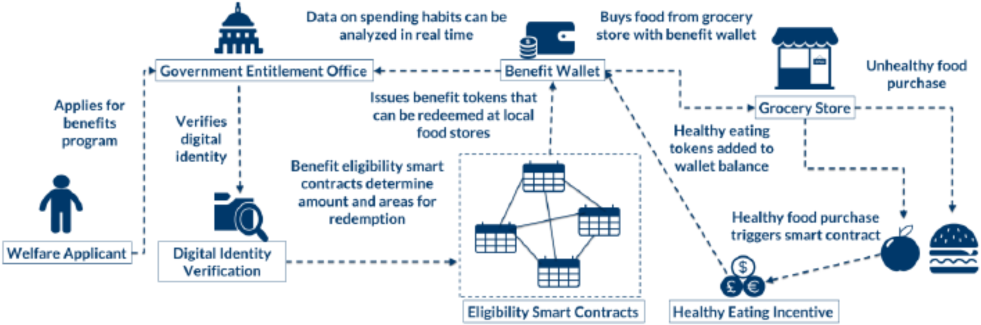
Smart contract solution: unified ledger connects government, employers, and providers; eligibility updates instantly when income changes; benefits auto-enroll or stop

Medicaid pilot: distributed ledger of eligible recipients; real-time income data; eliminates duplicate enrollment across states

WFP Building Blocks: blockchain for refugee food assistance in Jordan; eliminated 98% of bank transaction fees

SNAP/TANF	Student Loans	Housing Support
Medicaid/Medicare	Disaster Recovery Grants	Tax Collection
Unemployment Insurance	Research Grants	Municipal Grants
Worker's Compensation	Agricultural Price Support	Conservation Grants

Entitlement Digital Currencies and Programmable Money



Consumer, Retail, and Sharing Economy

Blockchain in Consumer and Retail Applications

Loyalty and Brand Engagement

- **Starbucks Odyssey (2023)**: blockchain loyalty platform on Polygon; customers earn and trade Journey Stamps (NFTs) for real-world rewards; combines gamification and digital collectibles
- **Nike .SWOOSH**: official Web3 platform for co-created digital sneakers on Polygon; smart contracts auto-pay royalties to co-creators; consumers become micro-entrepreneurs; $185M+ in NFT revenue
- **VeChain authenticity**: partners with Givenchy, Walmart China, and wine producers; luxury goods and food tagged with unique blockchain IDs for provenance verification

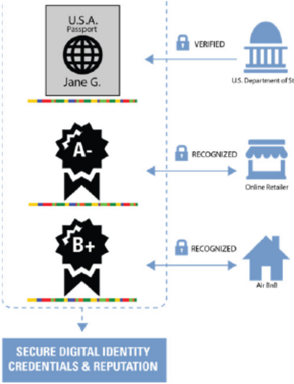
Connected Devices and Sharing Economy

- **Samsung Nexledger**: enterprise blockchain for IoT device management; smart meters transact autonomously (e.g., micropayments for power); reduces fraud in device networks
- **Sharing economy problem**: Airbnb and Booking charge 10–20% service fees; reviews can be manipulated; identity verification is slow
- **Blockchain solution**: DIDs verify host and guest credentials; smart contracts automate booking, deposit, and refund; on-chain ratings are tamper-proof; instant crypto payments upon check-in/out
- **Origin Protocol**: decentralized P2P marketplace; eliminates Airbnb intermediary; lower fees for hosts

Key Lesson

Nike and Starbucks show that blockchain consumer applications work when they add real utility — not just speculative value.

Blockchain Verification of Credentials and Reputation



Source: Goldman Sachs Global Investment Research.

Case Study: Sharing Economy in Lodging

The Sharing Economy: Lodging

$3-9bn increase in US booking fees through 2020

What blockchain can do

Ease identity and reputation management. Blockchain can securely store and integrate users' online transaction and review history with identification and payment credentials—making it easier to establish trust between parties. This information can be used to streamline transactions and enhance review quality.

Select enablers

Airbnb, HomeAway, FlipKey, OneFineStay

Incumbents at risk

Hotel industry

Blockchain Applications Landscape

Topic	Organization	Singular Idea
Accounting	The American Institute of CPAs, ConsenSys, Balanc3 (Deloitte, Ernst & Young, KPMG, and PwC consortia)	Triple-entry accounting on blockchain.
Automobile sales	Visa and DocuSign	Blockchain to build proof of concept for car leasing: click, sign, and drive.
Banking	R3 (consortium of more than 70 major banks)	Corda synchronizes financial agreements among members.
Cloud storage	Storj	Storj and Counterparty developing near-instantaneous bitcoin micropayments.
Cyber security	Guardtime and Enigma	Using blockchains to fight cyber attacks.
Education	Holbertson School, Sony Global Education	Recording students' results on blockchain.
Energy	LO3, ConsenSys	Paid energy trade using blockchain.
Finance—stocks	Nasdaq	Opening blockchain services to global exchange partners.

Source: Risk Block

Blockchain Applications: Insurance and Risk

Energy	LO3, ConsenSys	Paid energy trade using blockchain.
Finance—stocks	Nasdaq	Opening blockchain services to global exchange partners.
Forecasting	Augur	Blockchain prediction market enters beta testing.
Government	Japanese government	Japan sends blockchain start-ups abroad as part of innovation program.
Healthcare	IBM Watson, U.S. Food and Drug Administration	IBM Watson and FDA use blockchain to improve public health.
Internet of Things	Chronicled, Amazon	Chronicled launches Internet of Things registry.
Mass media entertainment	Disney	Disney develops its own blockchain: the Dragonchain.
Money transfers	SWIFT	SWIFT testing blockchain technology.
Music	PledgeMusic, PeerTracks, and BitTunes	Using blockchain technology to change the music industry.
Real estate	Propy	Using blockchain for local and international real estate deals.

Source: Risk Block

Blockchain Applications: Healthcare and Identity

Social media	Steemit	Steemit uses blockchain to create new social media network that pays for content.
Sports	Microsoft and BraveLog	Microsoft Azure develops first sports blockchain: BraveLog.
Supply chain management	Walmart	Walmart tests supply chain management using blockchain.
Voting	Expanse Borderless	Americans voting on blockchain.

Source: Risk Block

Synthesis: Where Blockchain Creates Value

Global Blockchain Adoption: Where We Stand

Market Metrics (2024)

- **Global spending**: \$19B in enterprise blockchain in 2024 (IDC); growing 40% annually
- **Financial institutions**: 80% have active blockchain initiatives (PwC, 2024)
- **Fortune 500**: 50%+ piloting blockchain in supply chain or finance
- **DeFi TVL**: \$90B+ recovering after 2022 crash; \$3.8B stolen in 2022 hacks highlights remaining risk
- **RWA tokenization**: \$8B+ on-chain; \$16T projected by 2030 (BCG)

Where Blockchain Wins

- **Wins**: cross-border payments (speed and cost), trade finance documentation, food traceability, tokenized securities settlement, digital identity
- **Struggles**: high-frequency trading (too slow), mass consumer adoption (UX too complex), governance (TradeLens, DAO attacks)
- **Emerging**: AI plus blockchain — AI agents executing smart contracts autonomously; blockchain as tamper-proof audit trail for AI model provenance

Key Lesson

WEF predicts 10% of global GDP on blockchain by 2030. The winners will be applications where trust, transparency, and automation create clear economic value.

Blockchain vs. Traditional Databases: When to Use Which

Use Blockchain When:

- **Multiple untrusting parties** need to share and update the same data (e.g., banks in a consortium, supply chain participants)
- **Immutability matters**: audit trail must be tamper-proof and verifiable by outsiders
- **No central authority**: no single party should control the ledger
- **Programmable settlement**: payments and delivery must be atomically linked
- **Example**: cross-border payment with 5 correspondent banks, 3 currencies, and regulatory reporting

Use a Traditional Database When:

- **Single organization**: one company controls and trusts all data
- **High throughput**: millions of transactions per second required (blockchain max ~15–100k TPS for Layer 2)
- **Data must be editable**: GDPR "right to be forgotten" conflicts with immutability
- **Cost matters**: blockchain adds overhead; SQL is cheaper for internal records

Key Lesson

The question is never "blockchain vs. database" in the abstract — it is whether the use case requires trust across untrusting parties at scale.

Discussion: Blockchain's Future in Finance

Open Questions

- Can **DeFi** ever achieve regulatory compliance without sacrificing decentralization?
- Will **CBDCs** displace private stablecoins, or will both coexist serving different purposes?
- Can **tokenized RWA** scale to $16T by 2030, or will legal and custody barriers slow adoption?
- Does blockchain **complement or replace** traditional financial intermediaries in the long run?

Strategic Implications

- **For banks**: blockchain reduces settlement costs but also reduces fee income from intermediation — creative destruction of their own business model
- **For regulators**: code is law in DeFi — enforcement requires understanding protocol design, not just reading contracts
- **For investors**: DeFi yields are real but so is smart contract risk; due diligence requires code audits, not just financial statements
- **AI synergy**: AI agents using blockchain smart contracts for autonomous transactions is the next frontier

Key Lesson

Blockchain is infrastructure, not a product. Like the internet, it will transform industries gradually — the biggest applications may not yet exist.